

Venkatesh Tantravahi

Utica - NY

+1-586-742-1300 ✉ vtantravahi@gmail.com <https://www.linkedin.com/in/venkateshtantravahi>

<https://venkateshtantravahi.netlify.app>

Summary

Data Scientist and Software Engineer with 2+ years of experience in machine learning, data analytics, and full-stack development. Expertise in Python, SQL, Power BI, and cloud platforms (AWS, Azure), with a strong focus on AI-driven solutions, data visualization, and business intelligence. Proven track record in computer vision models, anomaly detection, and data pipeline optimization.

SKILLS

- **Programming Languages:** Python, SQL, Go
- **Web Development:** Django, Flask, FastAPI, HTML, CSS, Bootstrap
- **Data Management:** MySQL, MongoDB, SQLite, SAP HANA
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Azure, Git
- **Data Science & Machine Learning:** TensorFlow, PyTorch, Keras, scikit-learn, Pandas, NumPy, statistics, Data Science, Machine Learning, Recommendation Systems, Causal Reasoning
- **Data Analytics & Visualization:** Tableau, SAS, Matplotlib, Seaborn, Plotly
- **Methodologies:** Agile, Scrum, Waterfall, STAR
- **Soft Skills:** Communication, Technological Adaptability, Problem Solving, Presentation

WORK EXPERIENCE

Capgemini

Jun 2021 - Dec 2022

Software Engineer

Hyderabad, India

- Developed Power BI dashboards to track sales performance by pulling data from SQL databases, reducing the time taken for manual report generation by 40% and increasing executive visibility into key KPIs.
- Led a team of two analysts to manage data requests, completing 95% of requests within the required turnaround time and improving stakeholder response efficiency.
- Optimized SQL queries used in sales reports by restructuring joins and indexing tables, decreasing query execution time by 8% and lowering database costs by 6%.
- Automated recurring reports by developing scheduled SQL jobs and Power BI refreshes, cutting manual reporting effort by 4 hours daily and reducing errors in reports.
- Implemented Row-Level Security (RLS) in Power BI to control data access based on user roles, ensuring compliance with company security policies and maintaining data privacy for over 500 users.
- Developed a data validation framework to identify inconsistencies in sales reports, reducing data discrepancies by 22% and improving decision-making accuracy.
- Integrated Power BI dashboards with external APIs to fetch live data updates, reducing data refresh delays by 50% and ensuring real-time insights for business teams.

Jukshio

Jul 2020 - Jun 2021

Data Scientist

Hyderabad, India

- Trained a facial recognition model using TensorFlow and Keras, applying quantization techniques to reduce model size by 25% while maintaining 98% recognition accuracy.
- Analyzed large-scale image datasets with Pandas and OpenCV, cleaning and preprocessing data to improve model training time by 30% and reduce false positives by 3%.
- Designed an anomaly detection system using statistical methods and scikit-learn, identifying inconsistencies in face embeddings and improving fraud detection rates by 12%.
- Built data pipelines to process real-time images using NumPy and OpenCV, reducing the inference time for model predictions by 20%, allowing faster recognition in production.
- Collaborated with engineers to deploy the model in a cloud environment, optimizing API response times by 15%, ensuring smooth integration with real-time security applications.
- Implemented a facial recognition API with Flask and TensorFlow Serving, enabling easy integration into multiple security applications and reducing deployment time by 35%.
- Designed a model evaluation system using LIME and SHAP to explain predictions, helping reduce incorrect classifications by 8% and increasing model transparency.

3IceCubes

May 2020 - Aug 2020

Full-Stack Developer Intern

Remote

- Developed a responsive web application with Django and MySQL, improving load time by 18% and enhancing user engagement by 7% through improved navigation.
- Automated data processing workflows in Python, reducing manual data entry errors by 20% and decreasing processing time by 10%.
- Optimized SQL queries in the backend, cutting query execution time by 12%, leading to faster data retrieval for users.

- Implemented a login authentication system with JWT tokens, securing user accounts and reducing unauthorized access incidents by 95%.
- Refactored frontend code in JavaScript and CSS, reducing page rendering time by 25% and improving responsiveness across different screen sizes.
- Integrated a REST API with the backend to handle user data updates dynamically, decreasing page reload times by 40% and improving user experience.
- Set up automated unit tests using Pytest for backend logic, catching 85% of critical bugs before deployment and reducing debugging time by 30%.

EDUCATION

State University of New York, Utica-NY	Jan 2023 - Dec 2024
Master of Science, Computer Science	
•GPA: 3.40/4.00	

PROJECTS

SQLite Playground	Mar 2024 - Apr 2024
• Designed and developed a web-based SQL editor that allows users to write, execute, and visualize SQL queries in an interactive environment. • Implemented syntax highlighting, query auto-completion, and real-time error detection to enhance the user experience. • Integrated data visualization features using Matplotlib and Plotly, enabling users to generate graphical insights from query results. • Built using Python and Streamlit, ensuring a lightweight and responsive interface for real-time SQL execution.	
Extending Forward-Forward Algorithm	Jan 2024 - Mar 2024
• Enhanced the Forward-Forward (FF) algorithm by adapting it to complex datasets and diverse activation functions, making it more effective for real-world applications. • Explored alternative learning approaches to backpropagation, leveraging advanced statistical techniques to improve convergence and stability. • Evaluated performance on benchmark datasets, identifying improvements in training efficiency and model generalization. • Conducted experiments on different activation functions and learning rates, assessing their impact on model performance.	
Rain in Australia	Dec 2023 - Dec 2023
• Conducted a climate trend analysis to predict rainfall patterns in Australia, using machine learning techniques. • Applied LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations) to interpret model predictions and identify key weather factors influencing rainfall. • Built a predictive model using scikit-learn to analyze historical climate data and improve forecasting accuracy. • Generated visual insights to illustrate how different environmental variables impact weather patterns, supporting data-driven decision-making.	